

MAP & Credible Intervals

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta; x)$$

↓
Data

↑
likelihood

Data is given, $L(\theta; x) = \{ \theta | x \}$

$\phi = h(\theta)$ h: one-one

$$\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$$

MLE is invariant for 1-1 transformation
i.e. $\hat{\phi}_{MLE} = h(\hat{\theta}_{MLE})$

Maximum A Posteriori Estimation (MAP)

$$\pi(\theta|x)$$

MAP is a point estimator of θ based on $\pi(\theta|x)$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} \pi(\theta|x)$$

mode of $\pi(\theta|x)$

$X \sim N(\mu, \sigma^2)$ mode is μ

$$Y = e^X, X = \log Y$$

e^{μ} (is the mode?)

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$f_Y(y) =$$

Mode of Y will be $e^{\mu - \sigma^2} \neq e^{\mu}$

$$L(\theta; x) \Leftrightarrow p(x|\theta)$$

$$\pi(\theta|x)$$

$L(\theta; x)$ (Likelihood function is not a prob. density fn.)

$$\int L(\theta; x) d\theta \neq 1 \quad \left[\int p(x|\theta) d\theta = 1 \right]$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} \pi(\theta|x) \quad \pi(\theta|x) \text{ is a prob. density fn.}$$

$$= \arg \max_{\theta} \left[L(\theta; x) \cdot \underbrace{\pi(\theta)}_{\text{prior}} \right] \quad \text{therefore MAP is not invariant to reparameterization.}$$

e.g. $X_1, \dots, X_n \sim \exp(\beta)$
 $\pi(\beta) \propto 1$. Then $\hat{\beta}_{MAP} = \frac{\sum X_i}{n}$

$$pdf = \frac{1}{\beta} \exp\left(-\frac{x}{\beta}\right), \hat{\beta}_{MLE} = \frac{\sum X_i}{n}$$

$$\pi(\beta|x) \propto \frac{1}{\beta^n} \exp\left(-\frac{\sum X_i}{\beta}\right) \Rightarrow \pi(\phi|x) \quad \log\left(\frac{\sum X_i}{n-1}\right)$$

$$\phi = \beta \log \beta$$

then find $\hat{\phi}_{MAP}, \hat{\phi}_{MLE}$

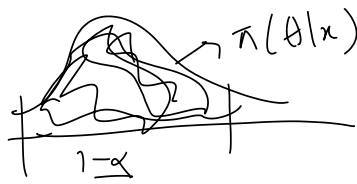
$$\hat{\phi}_{MLE} = \log\left(\frac{\sum X_i}{n}\right)$$

Posterior mean $\pi(\phi|x)$
mode (MAP)

Credible interval

$$P(L < \theta < U|x) = (1-\alpha)$$

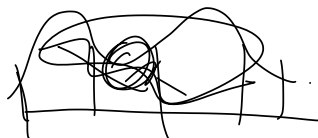
$$P(L < \theta < U) = 1-\alpha$$



e.g. $X|p \sim \text{Binomial}(n, p)$

$$p \sim \text{Beta}(a, b)$$

$$p|x \sim \text{Beta}(a+n, b+n-x)$$



$$\left(F^{-1}(0.025) \text{ Beta}(a+n, b+n-x), F^{-1}(0.975) \text{ Beta}(a+n, b+n-x) \right)$$

$X_1, \dots, X_n \sim N(\mu, \sigma^2)$ σ known.

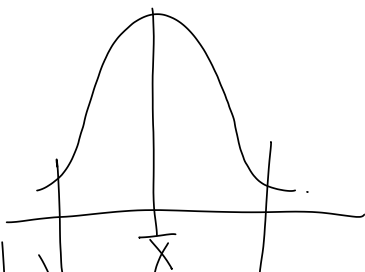
$$\pi(\mu) \propto 1$$

$$\text{Then, } \pi(\mu|x) \sim N\left(\bar{X}, \frac{\sigma^2}{n}\right)$$

$$\mu|x_1, \dots, x_n \sim N\left(\bar{X}, \frac{\sigma^2}{n}\right)$$

$$\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

$$\exp\left(-\frac{1}{2} \frac{(\ln \mu - \bar{X})^2}{\frac{\sigma^2}{n}}\right) \Rightarrow \left(\frac{1}{2} \frac{(\ln \mu - \bar{X})^2}{\frac{\sigma^2}{n}} + n(\bar{X} - \mu)^2 \right)$$



$$P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \mid X \right) = 0.95$$

$\hat{\theta}_{MLE}$

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) \sim N(0, I^{-1}(\theta_0))$$

$$\left(\hat{\theta} - 1.96 \frac{\sigma}{\sqrt{n}}, \hat{\theta} + 1.96 \frac{\sigma}{\sqrt{n}} \right) \quad \text{Asymptotic normality of MLE}$$

Bernstein von Mises theorem

$$\sqrt{n}(\hat{\theta} - \theta_0) \mid X \sim N(0, \sigma_0^2)$$

① Paradox of classical inference.

② Why Bayesian. How to do Bayesian (prior, likelihood $\Rightarrow$ posterior)

③

Prior

→ Conjugate prior (good for calculations).

Some concepts of non-informative prior. (Jeffreys).

prior, prior predictive distⁿ, posterior, posterior predictive distⁿ,
updating belief

→ Check at least all the cases mentioned in BS DS syllabus.

Properties of scaled-inverse- χ^2 distⁿ

→ unknown mean unknown variance & Dirichlet Multinomial

MAP & credible interval

Estimation: Prior over the parameter.
x Likelihood.

$$P(\theta|x)$$

MAP
Credible interval.

Bayesian hypothesis testing: Posterior odds, Bayes Factor

Diagnostic example: $P(D) = 0.02$ $P(T^+|D) = 0.95$
 $P(D^c) = 0.98$ $P(T^+|D^c) = 0.10$

$$P(D|T^+) = \frac{P(D)P(T^+|D)}{P(T^+)} = 0.1624$$

Posterior $(P(D|T^+)) \propto P(T^+|D)P(D)$

Hypothesis testing setup: $H_0: D^c$ $H_1: D$

$$P(H_0) = 0.98 \quad P(H_1) = 0.02$$

Compute posterior probabilities of both the hypotheses
Accept H_1 if $P(H_1|x) > 0.5$ [Decision Rule]

Posterior odds: $\frac{P(H_1|x)}{P(H_0|x)} = \frac{P(x|H_1)}{P(x|H_0)} \cdot \frac{P(H_1)}{P(H_0)}$

Posterior odds: $\frac{P(x|H_1)}{P(x|H_0)}$ Likelihood ratio

Prior odds: $\frac{P(H_1)}{P(H_0)}$

If $\frac{P(H_1|x)}{P(H_0|x)} > 1$, then accept H_1 .

Bayes Factor: $BF_{10} = \frac{P(x|H_1)}{P(x|H_0)}$ & then, $\frac{P(H_1|x)}{P(H_0|x)} = BF_{10} \frac{P(H_1)}{P(H_0)}$

$$BF_{10} = \frac{P(T^+|D)}{P(T^+|D^c)} = 9.5$$

Compatibility of the observed data with the hypotheses.

Posterior odds = BF_{10} (prior odds)
Bayes factor doesn't depend on the prior odds of the hypotheses.

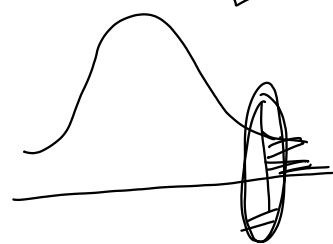




Composite hypotheses:

$H_0: \mu = 0$ vs $H_1: \mu \neq 0$
 $H_0: \mu \leq 0$ vs $H_1: \mu > 0$

Suppose. $X|A, H_j \sim f_j(x|A)$.
 under H_j , suppose $\pi_j(A)$ prior for θ .



$H_0: \mu = 0$ vs $H_1: \mu$
 $H_0: \mu \geq 0$ vs $H_1: \mu > 0$
 $\pi_1(\mu)$

$$P(x|H_j) = \int \underbrace{f_j(x|\theta)}_{\text{Likelihood}(\theta)} \underbrace{\pi_j(\theta)}_{\text{Likelihood}(\theta_2)} d\theta$$

$\frac{\text{Likelihood}(\theta_1)}{\text{Likelihood}(\theta_2)}$

(10) $H_0: X \sim N(\mu_0, 1)$ $\mu_0 \sim U(9, 11)$

$m_j(x) =$

(Marginal likelihood)

$f(10; 9, 2, 1) H_1: X \sim N(\mu, 1)$

(Prior predictive density)

$\mu \sim U(-9, 9)$
 $f(10; 1, 1)$

$f(10; -1, 1)$

$f(10; -3, 1)$

$H_1: X \sim N(\mu, 1)$
 $\mu \sim N(9, 1)$

$H_0: X \sim N(\mu, 1)$
 $\mu \sim N(-9, 1)$

D: 10

$BF_{10} > 1$

$BF_{10} = \frac{m_1(x)}{m_0(x)}$

Not using prior odds of the hypotheses.

$P(H_1|x)$

$P(H_0|x)$

$\propto \frac{P(x|H_1)P(H_1)}{P(x|H_0)P(H_0)}$
 Prior odds.

When prior odds are same,

posterior odds = $BF_{10} \left(\frac{m_1(x)}{m_0(x)} \right)$

$H_0: \theta = \theta_1$
 $H_1: \theta = \theta_2$

$\frac{m_1(x)}{m_0(x)}$

then $BF = \text{likelihood ratio}$

Rule of thumb.

if $P(H_1|x) > P(H_0|x)$ then accept H_1
 accept H_0

evidence = f(marginal likelihood)

if $\log_m(BF_{10}) \in (0, 0.5)$ the evidence against H_0 is

$$H_0: \mu = 0 \quad H_1: \mu \neq 0$$

$\mu = 0$ w.p. 1 $\mu \sim N(10, 1)$

$$\Sigma(0.5, 1)$$

$$\Sigma(1, 2)$$

$$p > 2$$

the ^{poor} evidence against H_0 is
substantial.
strong.
decisive

Note that,

$$\text{Likelihood} = \frac{f(x|A_1)}{f(x|A_2)}$$

$$\int f(x|A) dx = 1$$

$$\int f(x|\theta) dx = 1$$

$$\int f(x|A_2) dx = 1$$

Show this $\int m_1(x) dx = 1$

Likelihood ratio

$$\int f(x|\theta) d\theta \neq 1$$

$$\int f(x|\theta) dx = 1$$